1. Validate Data Accuracy

Scenario: Verify user data stored in DB matches UI input

Query:

```
SELECT * FROM users WHERE email='test@test.com';
```

Expected: All fields match UI.

2. Validate Foreign Key Relationship

Scenario: Every order must belong to a valid user

Query:

```
SELECT * FROM orders WHERE user_id NOT IN (SELECT user_id FROM users);
```

Expected: Zero records.

3. Validate Duplicate Records

Scenario: No duplicate emails allowed

Query:

```
SELECT email, COUNT(*) FROM users GROUP BY email HAVING COUNT(*) > 1;
```

Expected: Zero duplicates.

4. Validate NULL Restrictions

Scenario: email column should not allow NULL

Query:

```
SELECT * FROM users WHERE email IS NULL;
```

Expected: Zero records.

5. Validate Order Amount Calculation

Scenario: Order amount = sum of item prices

Query:

```
SELECT order_id, amount,
    (SELECT SUM(price) FROM order_items oi WHERE oi.order_id=o.order_id) AS calculated_amount
FROM orders o;
```

Expected: amount = calculated_amount.

6. Validate UPDATE Operation

Scenario: Update user email

Query:

UPDATE users SET email='new@test.com' WHERE user_id=10;

Expected: Email updated.

7. Validate DELETE Operation

Scenario: Delete user record

Query:

DELETE FROM users WHERE user_id=10;

Expected: Record removed.

📘 **8. Validate Index Performance**

Scenario: Query should use index on user_id

Query:

EXPLAIN SELECT * FROM orders WHERE user_id=5;

Expected: Index scan, not full table scan.